

FASAL 2.- November, 2024

Forecasts for Kharif Crops - 2024-25 States and All-India

25th November, 2024

FASAL TEAM

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COVERAGE

- Crop coverages - Rice, Bajra, Jowar, Maize, Arhar (Tur), Moong, Urad, Groundnut, Soybean, Castor Seed, Cotton, Jute, Sugarcane for Kharif season.
- Estimates are made at the level of major growing states depending on data availability and sample period is 2000-01 – 2022-23.
- **Rainfall and temperature are based on the information available for 24th November, 2024.**

Disclaimer

1. Normal rainfall and temperature assumed 25th November, 2024 onwards
2. We recently obtained groundwater data from CGWB for the first week of May 2024. However, we need data for the end of May, 2024 as a benchmark for the kharif forecasts. To address this, we've projected groundwater levels using an ARIMA model based on historical data. We anticipate receiving the latest data from CGWB soon.
3. **Irrigation data 2022 onwards, not available, projected from past data.**
4. Updated actual whole sale price of 2023 from DES and projected 2024 (Jan) onwards based on WPI (Office of the Economic Advisor, Website).
5. Forecasts based on model using only published data available. Actual realization may depend on contemporary field situation at special time.

Forecasts need to be cross-correlated with field insights.

FASAL 2. - November, 2024

Actual rainfall and temperature up to 24th November, 2024

Normal of all other future months of Kharif season and **MSP** of crops at **revised** levels of **2024-25**

(Units: Area: '000 Hectare, Yield: Kg/Hectare and Prod (Production): Million Tonnes)

SECTION- A

Forecasts for Kharif Crops in 2024-25

Table 1: Projected All India Kharif Crop Production – 2024-25									
	2024-25 – IEG Forecasts (November, 2024)				2024-25 (1AE) MoA &FW			2023-24 MoA &FW	
Crops	Area	Yield	Prod	Prod Interval	Area	Yield	Prod	Area	Prod
Rice Kharif	40756	2842	115.85	110.09 - 121.75	43368	2765	119.91	40734	113.26
Bajra	7105	1391	9.89	8.49 - 11.39	6746	1390	9.38	6966	9.66
Jowar Kharif	1557	1205	1.88	1.55 - 2.22	1919	1144	2.20	1289	1.51
Maize Kharif	8736	2845	24.86	22.47 - 27.34	8395	2923	24.54	8329	22.25
Arhar (Tur)	4643	895	4.15	3.72 - 4.60	4030	869	3.50	4131	3.42
Moong Kharif	3483	484	1.69	1.38 - 2.02	3309	418	1.38	3174	1.15
Urad Kharif	2796	603	1.68	1.36 - 2.04	2092	578	1.21	2681	1.60
Soybean	13299	1064	14.15	12.73 - 15.62	12935	1033	13.36	13255	13.06
GN Kharif	4327	2097	9.07	7.65 - 10.6	4913	2109	10.36	4044	8.66
Castor Seed	865	2015	1.74	1.53 - 1.97	787	1973	1.55	1041	1.96
Jute	599	2722	1.63 (9.1**)	1.56 - 1.70	526	2761	1.45 (8.1**)	598	1.66 (9.2**)
Cotton	12593	438	5.51 (32.4*)	4.95 - 6.10	11360	448	5.09 (29.9*)	12688	5.53 (32.5*)
Sugarcane	5439	80010	435.20	407.24 - 464.06	5369	81943	439.95	5740	453.16
Note: Projections are based on strong assumptions that the select states of estimation contribute the same share of all India area and production as the average of the same ratio of 2020-21, 2021-22 and 2022-23. * denoted million bales of 170 kgs. Each and ** denoted million bales of 180 kgs. each									

Table 2: Rice Kharif									
	2024-25 – IEG Forecasts (November, 2024)				2024-25 (1AE) MoA &FW			2023-24 MoA &FW	
States (Area share in India)	Area	Yield	Prod	Prod Interval	Area	Yield	Prod	Area	Prod
Andhra Pradesh (incl. Telangana) (9%)	3565	3512	12.52	11.91 - 13.14	3690	3440	12.69	3921	14.04
Assam (5%)	2005	2177	4.36	4.16 - 4.57	1802	2204	3.97	1918	4.37
Bihar (7%)	2843	2771	7.88	7.49 - 8.27	3139	2493	7.83	3050	7.73
Chhattisgarh (9%)	3851	2695	10.38	9.82 - 10.94	3835	2426	9.30	3822	9.70
Gujarat (2%)	901	2520	2.27	2.10 - 2.45	946	2534	2.40	873	2.14
Haryana (3%)	1565	3543	5.54	5.34 - 5.75	1644	3692	6.07	1563	5.97
Jharkhand (3%)	1357	1672	2.27	1.98 - 2.58	1381	1747	2.41	1100	1.52
Karnataka (2%)	954	3157	3.01	2.90 - 3.13	848	3255	2.76	849	2.82
Madhya Pradesh (8%)	3310	2378	7.87	7.34 - 8.42	3640	2114	7.69	3440	7.11
Maharashtra (4%)	1524	2181	3.32	3.20 - 3.45	1411	2437	3.44	1525	3.46
Orissa (9%)	3678	2157	7.93	7.40 - 8.47	3336	2032	6.78	3701	7.28
Punjab (7%)	3212	4337	13.93	13.47 - 14.4	3243	4527	14.68	3179	14.36
Tamilnadu (2%)	765	3027	2.32	2.03 - 2.62	657	3173	2.08	680	2.09
Uttar Pradesh (14%)	5728	2760	15.81	15.11 - 16.52	7976	2726	21.74	5732	15.87
Uttarakhand (1%)	238	2658	0.63	0.60 - 0.67	207	2488	0.52	221	0.56
West Bengal (10%)	4016	2883	11.58	11.23 - 11.93	4200	2875	12.08	3809	10.95
State Total (97%)	39512	2825	111.63	106.08 - 117.32	41955	2775	116.44	39383	109.97

Table 3. Bajra									
	2024-25 – IEG Forecasts (November, 2024)				2024-25 (1AE) MoA &FW			2023-24 MoA &FW	
States (Area share in India)	Area	Yield	Prod	Prod Interval	Area	Yield	Prod	Area	Prod
Gujarat (5%)	220	1829	0.40	0.27 - 0.54	227	1787	0.41	197	0.35
Haryana (7%)	557	2338	1.30	1.18 - 1.43	470	2299	1.08	528	1.16
Karnataka (2%)	137	1498	0.21	0.13 - 0.28	138	1290	0.18	128	0.16
Madhya Pradesh (5%)	350	2284	0.80	0.75 - 0.85	284	2574	0.73	388	1.00
Maharashtra (8%)	468	1150	0.54	0.42 - 0.67	399	794	0.32	367	0.21
Rajasthan (58%)	4284	1016	4.35	3.56 - 5.21	4227	1065	4.50	4257	4.38
Uttar Pradesh (13%)	982	2118	2.08	2.00 - 2.16	904	2182	1.97	1010	2.19
State Total (98%)	6998	1383	9.68	8.32 - 11.15	6649	1382	9.19	6875	9.47

Table 4. Jowar Kharif									
	2024-25 – IEG Forecasts (November, 2024)				2024-25 (1AE) MoA &FW			2023-24 MoA &FW	
States (Area share in India)	Area	Yield	Prod	Prod Interval	Area	Yield	Prod	Area	Prod
Karnataka (5%)	77	1846	0.14	0.11 - 0.18	60	1655	0.10	63	0.11
Madhya Pradesh (7%)	121	1816	0.22	0.16 - 0.29	90	1863	0.17	87	0.16
Maharashtra (16%)	105	954	0.10	0.02 - 0.19	108	916	0.10	112	0.10
Rajasthan (40%)	709	1062	0.75	0.70 - 0.81	605	977	0.59	509	0.45
Tamilnadu (13%)	163	773	0.13	0.10 - 0.15	167	970	0.16	159	0.16
Uttar Pradesh (12%)	285	1554	0.44	0.40 - 0.49	268	1592	0.43	289	0.46
State Total (94%)	1460	1222	1.78	1.48 - 2.11	1298	1191	1.55	1219	1.44

Table 5. Maize Kharif									
	2024-25 – IEG Forecasts (November, 2024)				2024-25 (1AE) MoA &FW			2023-24 MoA &FW	
States (Area share in India)	Area	Yield	Prod	Prod Interval	Area	Yield	Prod	Area	Prod
Andhra Pradesh (incl. Telangana) (4%)	470	4464	2.10	1.91 - 2.29	363	4025	1.46	355	1.50
Bihar (2%)	237	2692	0.64	0.56 - 0.72	190	2174	0.41	216	0.59
Chhattisgarh (2%)	112	2953	0.33	0.30 - 0.36	117	3333	0.39	109	0.34
Gujarat (4%)	301	1936	0.58	0.52 - 0.65	305	2022	0.62	282	0.57
Karnataka (19%)	1870	2920	5.46	5.07 - 5.87	1545	2980	4.60	1745	4.75
Jharkhand (3%)	223	2395	0.53	0.50 - 0.57	253	2088	0.53	259	0.54
Madhya Pradesh (18%)	1620	3092	5.01	4.60 - 5.43	2080	3084	6.41	1530	4.27
Maharashtra (11%)	838	2203	1.85	1.59 - 2.11	1107	2523	2.79	909	1.44
Punjab (1%)	99	3820	0.38	0.35 - 0.41	86	3946	0.34	95	0.36
Rajasthan (12%)	926	2386	2.21	2.02 - 2.40	949	2402	2.28	862	2.06
Tamilnadu (3%)	191	6995	1.34	1.03 - 1.66	228	7688	1.75	240	1.75
Uttar Pradesh (9%)	811	2298	1.86	1.70 - 2.03	287	2333	0.67	839	1.83
State Total (88%)	7698	2895	22.29	20.15 - 24.51	7510	2964	22.26	7441	20.01

Table 6. Arhar (Tur)									
	2024-25 – IEG Forecasts (November, 2024)				2024-25 (1AE) MoA &FW			2023-24 MoA &FW	
States (Area share in India)	Area	Yield	Prod	Prod Interval	Area	Yield	Prod	Area	Prod
Andhra Pradesh (incl. Telangana) (12%)	525	615	0.32	0.30 - 0.35	495	535	0.27	412	0.24
Gujarat (5%)	236	1318	0.31	0.29 - 0.33	173	1196	0.21	227	0.26
Jharkhand (5%)	287	1268	0.36	0.34 - 0.39	216	1139	0.25	189	0.22
Karnataka (33%)	1534	711	1.09	0.96 - 1.23	1560	785	1.22	1365	1.02
Madhya Pradesh (4%)	240	1082	0.26	0.20 - 0.32	144	882	0.13	155	0.14
Maharashtra (28%)	1170	927	1.08	0.99 - 1.19	1210	983	1.19	1112	0.89
Orissa (3%)	139	1167	0.16	0.15 - 0.18	15	1054	0.02	134	0.14
Uttar Pradesh (7%)	376	1131	0.43	0.39 - 0.47	84	1164	0.10	424	0.40
State Total (97%)	4507	892	4.02	3.60 - 4.46	3897	865	3.37	4018	3.31

Table 7: Moong Kharif									
	2024-25 – IEG Forecasts (November, 2024)				2024-25 (1AE) MoA &FW			2023-24 MoA &FW	
States (Area share in India)	Area	Yield	Prod	Prod Interval	Area	Yield	Prod	Area	Prod
Gujarat (2%)	97	485	0.05	0.04 - 0.06	29	554	0.02	65	0.04
Karnataka (12%)	541	328	0.18	0.15 - 0.21	433	354	0.15	283	0.09
Madhya Pradesh (2%)	14	556	0.01	0.01 - 0.01	45	560	0.03	26	0.01
Maharashtra (9%)	186	460	0.09	0.07 - 0.10	232	501	0.12	185	0.07
Rajasthan (67%)	2369	516	1.22	0.99 - 1.47	2396	399	0.96	2364	0.81
State Total (92%)	3207	480	1.54	1.26 - 1.85	3135	404	1.27	2923	1.03

Table 8: Urad Kharif									
	2024-25 – IEG Forecasts (November, 2024)				2024-25 (1AE) MoA &FW			2023-24 MoA &FW	
States (Area share in India)	Area	Yield	Prod	Prod Interval	Area	Yield	Prod	Area	Prod
Gujarat (4%)	125	679	0.08	0.07 - 0.10	85	619	0.05	79	0.05
Jharkhand (4%)	142	1063	0.15	0.14 - 0.17	115	882	0.10	105	0.09
Karnataka (2%)	83	661	0.05	0.05 - 0.06	115	882	0.10	44	0.02
Madhya Pradesh (41%)	874	608	0.53	0.41 - 0.66	433	574	0.25	797	0.50
Maharashtra (12%)	323	495	0.16	0.13 - 0.19	389	512	0.20	260	0.10
Odissa (2%)	99	398	0.04	0.03 - 0.04	17	415	0.01	83	0.03
Rajasthan (12%)	311	503	0.16	0.12 - 0.20	298	422	0.13	296	0.15
Uttar Pradesh (16%)	608	534	0.32	0.26 - 0.39	364	622	0.23	711	0.44
State Total (92%)	2565	586	1.50	1.22 - 1.82	1816	585	1.06	2375	1.39

Table 9. Soybean									
	2024-25 – IEG Forecasts (November, 2024)				2024-25 (1AE) MoA &FW			2023-24 MoA &FW	
States (Area share in India)	Area	Yield	Prod	Prod Interval	Area	Yield	Prod	Area	Prod
Gujarat (2%)	261	1440	0.38	0.33 - 0.42	293	1555	0.46	266	0.37
Karnataka (3%)	413	921	0.38	0.33 - 0.43	414	1077	0.45	389	0.41
Madhya Pradesh (48%)	5902	1084	6.40	5.86 - 6.96	5777	915	5.29	6060	5.47
Maharashtra (36%)	5120	1034	5.29	4.79 - 5.81	5052	1119	5.65	5115	5.26
Rajasthan (9%)	1266	1002	1.27	1.03 - 1.52	1134	1038	1.18	1126	1.17
State Total (97%)	12962	1058	13.72	12.35 - 15.14	12670	1027	13.02	12956	12.68

Table 10. Groundnut Kharif									
	2024-25 – IEG Forecasts (November, 2024)				2024-25 (1AE) MoA &FW			2023-24 MoA &FW	
States (Area share in India)	Area	Yield	Prod	Prod Interval	Area	Yield	Prod	Area	Prod
Andhra Pradesh (incl. Telangana) (14%)	247	1136	0.28	0.13 - 0.46	288	646	0.19	252	0.19
Gujarat (40%)	1868	2474	4.62	3.70 - 5.60	1895	2757	5.22	1635	4.51
Karnataka (8%)	321	669	0.21	0.17 - 0.27	326	708	0.23	259	0.15
Madhya Pradesh (7%)	459	2019	0.93	0.86 - 1.00	697	2060	1.44	468	0.97
Maharashtra (4%)	161	1329	0.21	0.20 - 0.23	147	1146	0.17	145	0.16
Rajasthan (17%)	904	2321	2.10	1.97 - 2.23	904	2353	2.13	884	2.02
Tamilnadu (5%)	203	2324	0.47	0.42 - 0.53	205	2174	0.45	197	0.42
State Total (96%)	4163	2120	8.83	7.45 - 10.31	4462	2200	9.82	3840	8.40

Table 11. Castor Seed									
	2024-25 – IEG Forecasts (November, 2024)				2024-25 (1AE) MoA &FW			2023-24 MoA &FW	
States (Area share in India)	Area	Yield	Prod	Prod Interval	Area	Yield	Prod	Area	Prod
Gujarat (74%)	542	2332	1.26	1.1 - 1.43	560	2270	1.27	725	1.60
Rajasthan (20%)	269	1691	0.45	0.4 - 0.51	173	1514	0.26	223	0.33
State Total (94%)	811	2119	1.72	1.5 - 1.94	733	2092	1.53	948	1.92

Table 12. Cotton									
	2024-25 – IEG Forecasts (November, 2024)				2024-25 (1AE) MoA &FW			2023-24 MoA &FW	
States (Area share in India)	Area	Yield	Prod	Prod Interval	Area	Yield	Prod	Area	Prod
Andhra Pradesh (incl. Telangana) (21%)	2474	460	1.14	1.06 - 1.21	2235	444	0.99	2240	0.99
Gujarat (18%)	2424	598	1.45	1.28 - 1.62	2392	597	1.43	2683	1.54
Haryana (5%)	537	579	0.31	0.28 - 0.34	476	365	0.17	578	0.26
Karnataka (6%)	686	456	0.31	0.27 - 0.35	667	473	0.32	743	0.35
Madhya Pradesh (5%)	646	457	0.30	0.28 - 0.31	512	469	0.24	630	0.31
Maharashtra (34%)	4485	298	1.34	1.19 - 1.49	4084	353	1.44	4234	1.37
Orissa (2%)	220	562	0.12	0.12 - 0.13	147	544	0.08	216	0.12
Punjab (2%)	125	648	0.08	0.05 - 0.12	100	409	0.04	214	0.11
Rajasthan (6%)	668	613	0.41	0.36 - 0.47	662	524	0.35	1004	0.45
Tamilnadu (1%)	139	369	0.05	0.05 - 0.05	56	331	0.02	130	0.04
State Total (99.5%)	12404	444	5.51	4.94 - 6.10	11331	448	5.08	12672	5.52

Table 13. Jute									
	2024-25 – IEG Forecasts (November, 2024)				2024-25 (1AE) MoA &FW			2023-24 MoA &FW	
States (Area share in India)	Area	Yield	Prod	Prod Interval	Area	Yield	Prod	Area	Prod
Assam (10%)	55	2221	0.12	0.11 - 0.13	58	2158	0.13	57	0.12
Bihar (8%)	52	2367	0.12	0.11 - 0.13	51	2511	0.13	53	0.14
West Bengal (81%)	482	2833	1.37	1.31 - 1.42	410	2894	1.19	479	1.39
State Total (98%)	590	2734	1.61	1.54 - 1.68	519	2774	1.44	589	1.65

Table 14. Sugarcane									
	2024-25 – IEG Forecasts (November, 2024)				2024-25 (1AE) MoA &FW			2023-24 MoA &FW	
States (Area share in India)	Area	Yield	Prod	Prod Interval	Area	Yield	Prod	Area	Prod
Andhra Pradesh (incl. Telangana) (1%)	99	87163	8.63	7.45 - 9.84	42	85381	3.59	54	4.65
Bihar (4%)	215	62020	13.33	12.53 - 14.16	208	59789	12.44	202	12.07
Gujarat (4%)	203	75746	15.40	14.78 - 16.03	240	76304	18.31	202	14.47
Haryana (2%)	96	83004	7.98	7.52 - 8.46	102	81696	8.33	88	7.36
Karnataka (11%)	493	89620	44.18	41.6 - 46.83	466	87000	40.54	543	41.81
Madhya Pradesh (2%)	106	71239	7.55	6.75 - 8.39	107	69322	7.42	115	7.59
Maharashtra (24%)	1452	77824	113.00	103.12 - 123.34	1364	86774	118.36	1437	112.09
Punjab (2%)	74	93718	6.94	6.20 - 7.70	95	84114	7.99	90	7.49
Tamilnadu (3%)	149	111534	16.62	14.4 - 18.89	101	107000	10.81	152	15.96
Uttar Pradesh (45%)	2389	80326	191.90	183.84 - 200.11	2442	81214	198.32	2653	215.81
State Total (97%)	5277	80647	425.53	398.19 - 453.75	5167	82468	426.11	5536	439.30

SECTION – B (Model and Methodology)

Equations of Hybrid Model

Area Equation for Crop

$$A_t = a_0 + a_1 A_{t-1} + a_2 P_t^C + a_3 \sum R_{tMm} + a_4 \sum IRG_{ts} + a_5 \sum RV_{t0} + a_6 \sum GD_{tM0} + a_7 \sum Z_{tz} \dots\dots\dots (1)$$

Yield Equation for Crop

$$Y_t = b_0 + b_1 t + b_2 P_t^C + b_3 P_t^S + b_4 \sum R_{tMm} + b_5 \sum I_{ts} + b_6 \sum TX_{tMf} + b_7 \sum TN_{tMf} + b_8 \sum RV_{t0} + b_9 \sum GD_{tM0} + b_{10} D_{tk} + b_{11} \sum Z_{tz} \dots\dots\dots (2)$$

Production of a Crop

$$P_t = A_t \times Y_t \dots\dots\dots (3)$$

A_t = area in hectares under study crop, Y_t = yield (Kg) per hectare of study crop, $P_t = P_{t1}$ or P_{t2} (equation 1) and $p_t = p_{t1}$ or p_{t2} (Equation 2) in alternate specifications where $P_{t1} = (EP_{tM}) / (EP_{tM}^S)$, $P_{t2} = (EP_{tM}^C * Y_{t-1}) / (EP_{tM}^S * Y_{t-1})$, $p_{t1} = (EP_{tM}^C) / (EP_{tM}^S)$, $p_{t2} = (EP_{tM}^C * Y_{t-1}) / (EP_{tM}^S)$, $p_{t1}^S = (EP_{tM}^C * Y_{t-1}) / (EP_{tM}^S)$, $p_{t2}^S = (EP_{tM}^C * Y_{t-1}) / (EP_{tM}^S)$.

EP_t (Expected Price) is the corresponding wholesale price (WSP) of crop in previous year, or latest MSP or the average of both, $I_{ts} = IRG_{ts} / \text{Exp}(A_t)$. Where R_{tMm} = rainfall averaged across over different alternate sets of M and m, IRG_{ts} is command area under any irrigation sources, TX_{tMf} = maximum of daily temperatures or their average across M, m and f of growing season, TN_{tMf} = minimum of daily temperatures or their average across M, m and f of growing season, RV_{t0} = Reservoir Volume for benchmark month (m=0), GD_{tM0} = Ground Water level for benchmark months (m=0) of met-regions of the study state, Interactions of rainfall (R_{tMm}) with irrigation ($R_{tMm} * I_{ts}$), and of irrigation and rainfall with reservoir ($RV_{t0} * I_{ts}$ and $R_{tMm} * RV_{t0}$) and ground water levels ($R_{tMm} * GD_{tM0}$ and $R_{tMm} * GD_{tM0}$) are also allowed as variables. D_k = kth dummy variable for any known change in technology or policy, Z_{tz} = any other relevant zth variable

Subscripts: t is year (2000-01 onwards), M=met region (1, 2,...34), m= months (April-March), f= fortnight (1, 2,...24), s= source of irrigation (Canal, well and tank and others). 0 = Benchmark month, Superscripts: C= Study crops, S= Substitute or competing crop (1, 2, . . . n), F= input fertilizer etc., exp = expected value of crop area = area estimated in equation (1).

I. Estimation

1. Equation specifications are selected on the basis of OLS equations and their diagnostics. Seemingly unrelated regression equation (SURE) Method applied for final estimation and forecasts.
2. The equations used for area are a function of crop price (expected), substitute crop price (expected), irrigation, lagged area, pre-sowing month reservoir and groundwater level data, rainfall in *sowing* season and fertilizer price. The equations for yield are expressed as a function of crop price (expected), substitute crop price (expected), irrigation, pre-sowing month reservoir and groundwater level, sowing season rainfall, rainfall in *growing* season, temperature in *growing season* and input price- fertilizer, Nutrients (DAP, Ammonium Sulfate), Pesticides, Insecticides price.

II. Water Specification

Water supplied for cultivation is obtained from natural rainfall and by irrigation drawing from surface (canal and tank) and ground water (well) sources using different technology but in turn irrigation also depends on rainfall, current and in the past and also in other regions especially upstream regions. Rain water is brought down as per gradient by river and some of the water seeps into the ground to sub-soil level. Other water remains as soil moisture or surface water and gradually gets evaporated unless taken in by plants. Past accumulated rainfall water is therefore stored either as natural ground water reserves or in man-made reservoirs where dams are built on the rivers by the government for water use in future seasons. At the local level, rainfall is stored in tanks. Ground water level and reservoir level are therefore measures of water available from past rainfall.

i. Reservoir Level

Data on water level in 127 reservoirs across the country as provided by CWC has been used. Generally, for kharif crop water level of June month end and for rabi crop, water level of November end are used as variables and the reservoirs of relevance are identified based on location, river hydrology and water sharing system in relation to the study states but the work is in progress and will be further refined.

ii. Ground Water Level

Ground water data for district level as provided by CGWB have been summarized by averaging (weighted by NSA) into meteorological (MET) region wise ground water level which is used as a variable to explained area and yield of crops of the state containing the respective METs region. Generally, for kharif crop, water level of end of May is used as benchmark month.

iii. Rainfall Specification

In reality the rainfall distribution across different met sub-divisions in the state can be important because cropping pattern is not uniform and some crops are concentrated in certain sub-state regions.

Production response can be sensitive to the location. The effect of rainfall in certain regions June be stronger, weaker or absent as compared to others. Also rainfall in regions even outside the state can be important for the state's agriculture through the flows in rivers and canals, reservoir storage, water sharing agreements and seepage.

Rainfall at the meteorological sub-divisions, (in total of 36) are used to explain acreage and the yield of crops keeping in context the meteorological map of the country and cropping pattern across each state. Rainfall of the pre-sowing month and entire possible growing season as comprehended from discussion about crop calendar have been considered for capturing soil moisture and seasonal water availability qualified with irrigation facility as available. Past rainfall which June also be available is subsumed in the reservoir and ground water variable.

III. Dummy Variables

1. Dummy for National food Security Mission (NFSM) is used for post 2007 years in case of maize and pulses. Observed error behavior suggests this use.

IV. Dependent and expected values required for computing explanatory variable forecasts and data sources

1. State-wise Acreage and Yield (Historical) – DES, Website, MoA&FW
2. Price (alternately revenue) of crop, substitute crops – Agriculture Price in India (Various) – DES, MoA&FW
3. Minimum Support Price - DES, Website, MoA&FW
4. Price of fertilizers, Pesticides (wholesale price index)- Office Of The Economic Adviser, Department For Promotion Of Industry And Internal Trade

5. Wholesale price index all commodities (WSPIA)- Office Of The Economic Adviser, Department For Promotion Of Industry And Internal Trade
6. Irrigated area (source wise and total)- DES, Website, MoA&FW
7. Met Region Rainfall (Monthly) – India Meteorological Department (IMD)
8. Temperature (Fortnightly) - India Meteorological Department (IMD)
9. Reservoir Level (Month end) – Central Water Commission, Ministry of Jal Shakti, GoI.
10. Ground Water Level (Month end)– Central Ground Water Board
11. Agriculture Wage rate (Ploughing/Sowing/Harvesting)– Monthly- Trial
12. Consumer price indices for Agriculture – Monthly- Trial

V. Projection of expected values for the forecast year

1. Forecasts of irrigated area, WSPIA (wholesale price index all commodities), and fertilizers price (price Index) are all based on distributed lag model.

$$\Delta y_t = a_0 + \sum_k \Delta y_{t-k}, \text{ using 3 lags, } k=3.$$

Where, y_t is the current period variable to be projected

2. Normal Rainfall has been used in all the forecasts. Normal rainfall for any month is the average of the normal rainfall of that month as reported (IMD) for the three years: 2021, 2022 and 2023. Forecasts is made assuming temperature (both maximum and minimum) also will be normal.

VI. Forecast adjustment

Forecasts are made only for major producing states using our model. The share of the included states in the area and production of India is approximated by the average of the past three consecutive years (2020-21, 2021-22 and 2022-23). The projections for 'All-India' are made using the totals of the forecast states and the projected share of the states assuming that this share holds true for the forecast year. Crop yields are then projected for 'all India' using these estimates of area and production at the 'all India' level.